



VLSI Testing

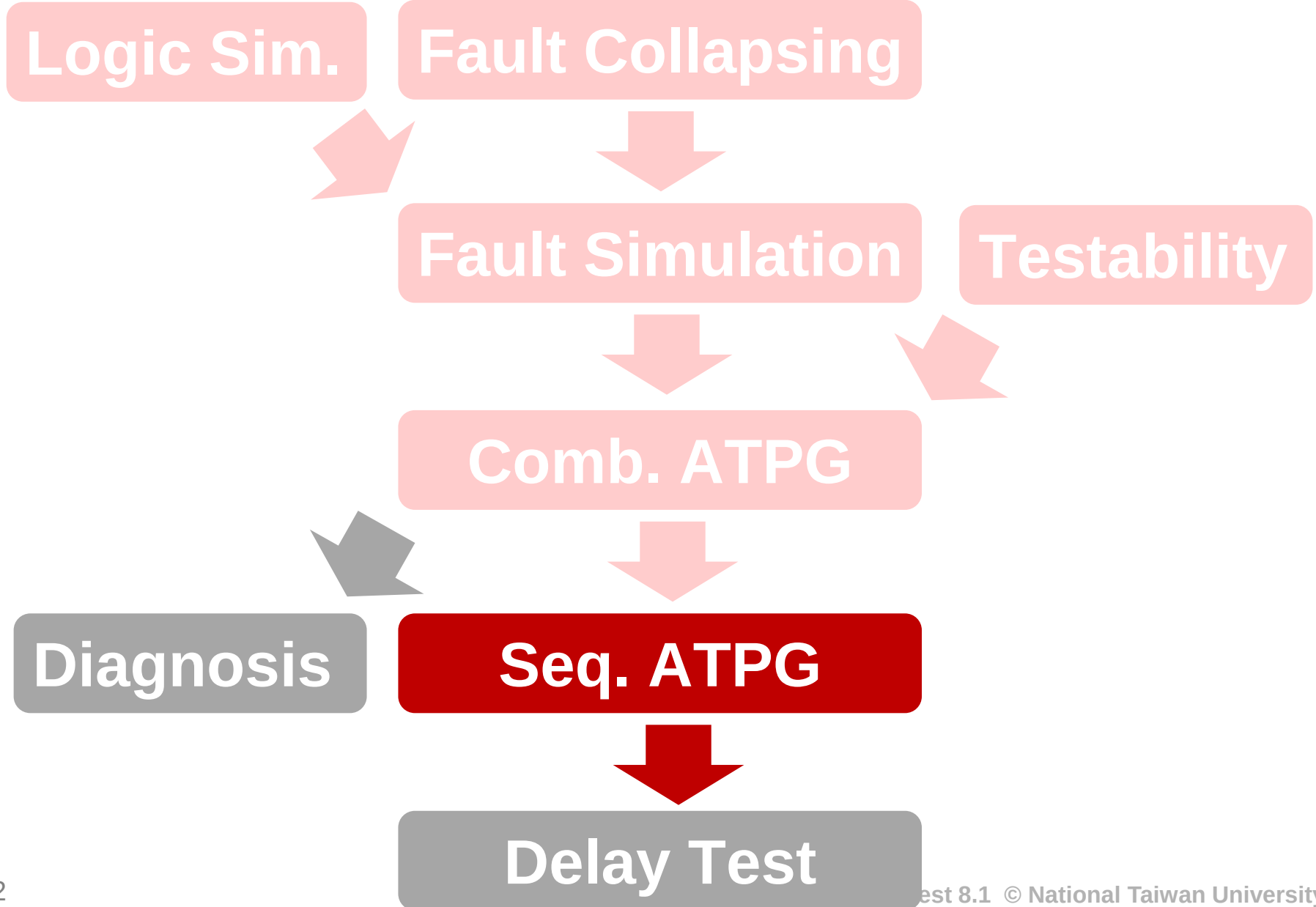
積體電路測試

Sequential ATPG

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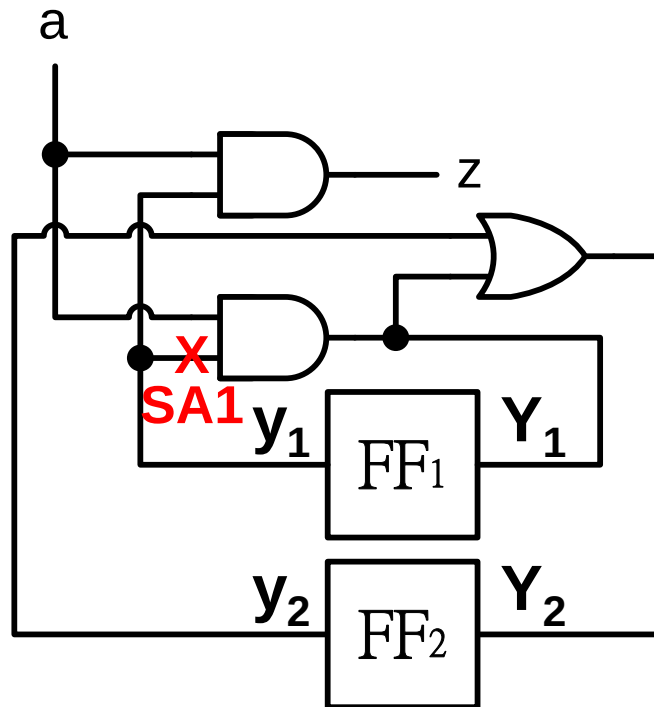
* Some picture are courtesy of Prof. Jiun-Lang Huang, NTU

Course Roadmap (EDA Topics)



Motivating Problem

- You already know ATPG for combinational circuits
- But manager asks you to generate a test for sequential circuits
 - ◆ No scan allowed in flip-flops (FF)



Why Am I Learning This?

- Sequential ATPG
 - ◆ Generate test patterns for sequential circuits
 - ◆ Without DFT or scan

***“If life is a test,
one should wish it to be short.”***

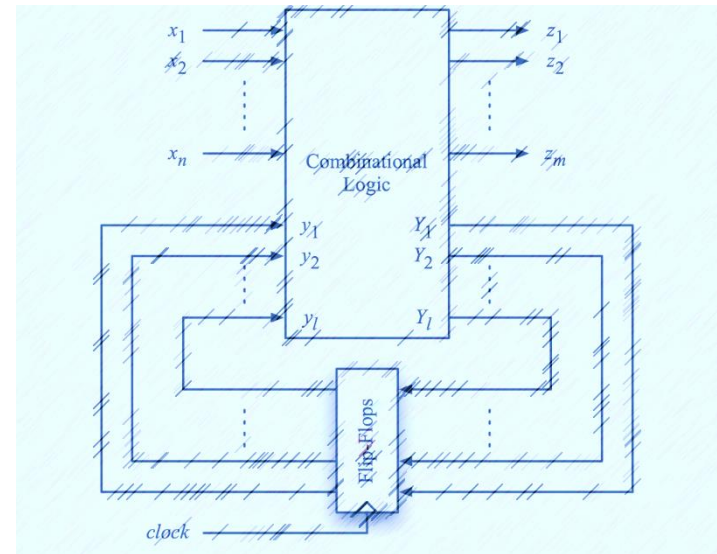
(Bernardin de Saint-Pierre)

Test Generation

Fault Models	Combinational Circuits (or Sequential ckt. with scan)	Sequential Circuits
No fault model	PET	Checking experiment
Single Stuck-at Fault Model	D PODEM FAN	Extended D 9-valued
Delay Fault Model	Path Delay Transition Delay Fault	Launch on Capture Launch on Shift

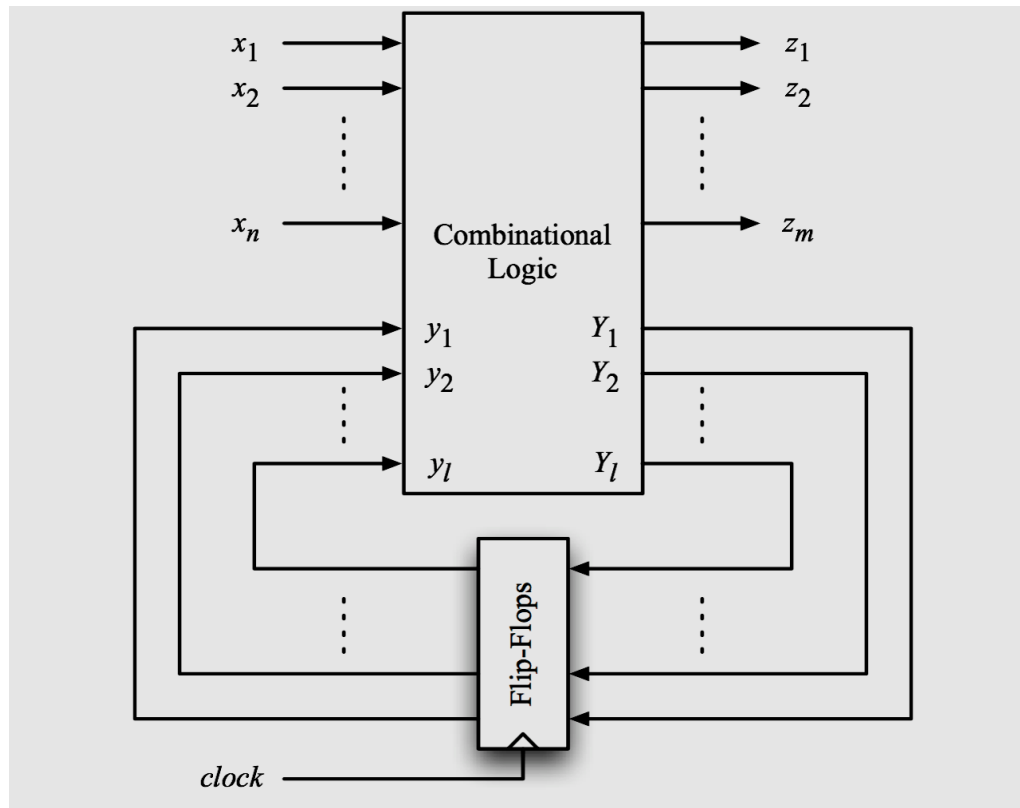
Sequential ATPG

- Introduction
- Time-frame expansion methods
- Simulation-based methods * (not in exam)
- Issues of Sequential ATPG * (not in exam)
- Conclusions



Huffman Model for Sequential Ckt. [Huffman 53]

- x_i = primary inputs (PI)
- z_i = primary outputs (PO)
- y_i = FF (or latch) next states
- Y_i = FF (or latch) current states

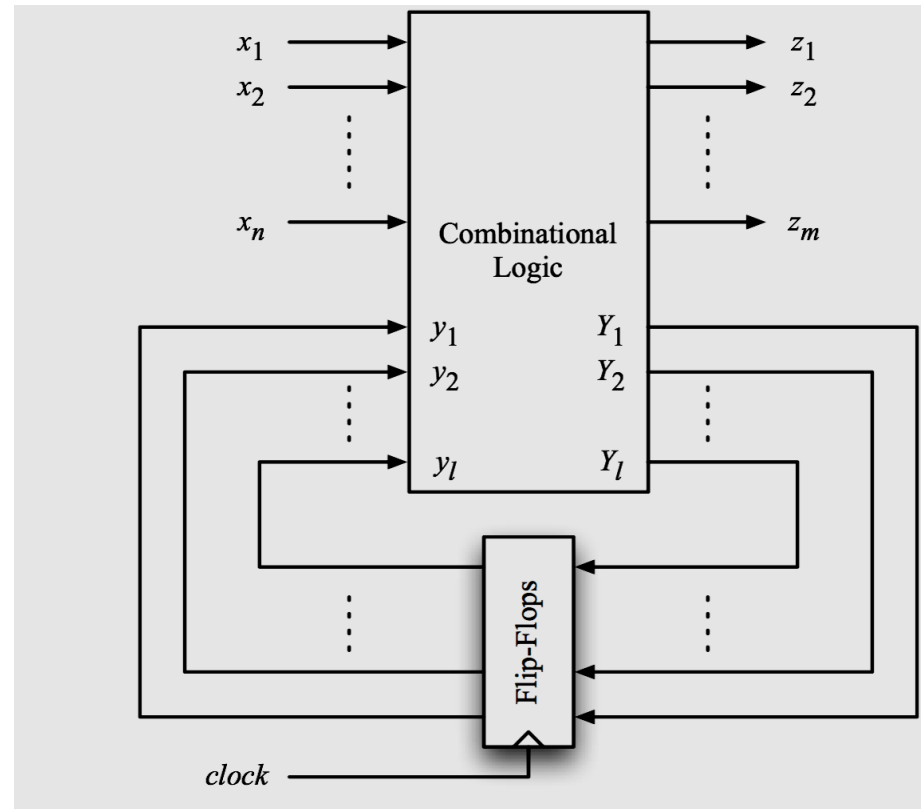


(WWW Fig 3.3)

Can be either
Moore or Mealy

Sequential ATPG Assumptions

- **NO SCAN ALLOWED!**
 - ◆ Control only PI
 - FF not controllable
 - ◆ Observe only PO
 - FF not observable
- **Faults in CL only**
 - ◆ NO fault in FF/latches



Challenges of Sequential ATPG

- 1. FF/latches states uncontrollable and unobservable
 - ◆ FF/latch unknown initial states
- 2. Long run time
 - ◆ Comb. ATPG complexity (for a given fault)
 - $O(2^{\text{number_of_PI}})$
 - ◆ Seq. ATPG complexity (for a given fault)
 - $O(2^{\text{number_of_PI}} \times 9^{\text{number_of_FF}})$
- 3. Large memory space required
 - ◆ Time frame expansion
- 4. Low fault coverage
 - ◆ Much worse than Comb. ATPG

*9-valued logic will be covered soon

Seq. ATPG is More Difficult Than Comb. ATPG

Quiz

Q: For a sequential circuit with 100 flip-flops, what is worst case sequential ATPG complexity?

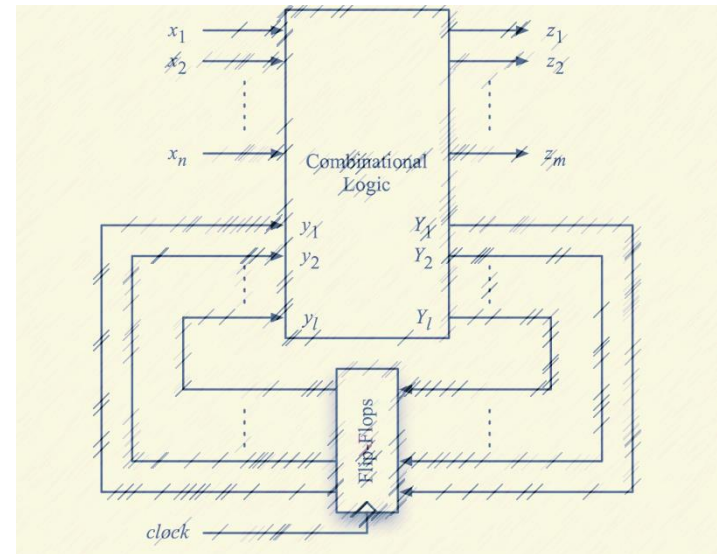
A: 9^{100}

B: 2^{100}

C: 100^2

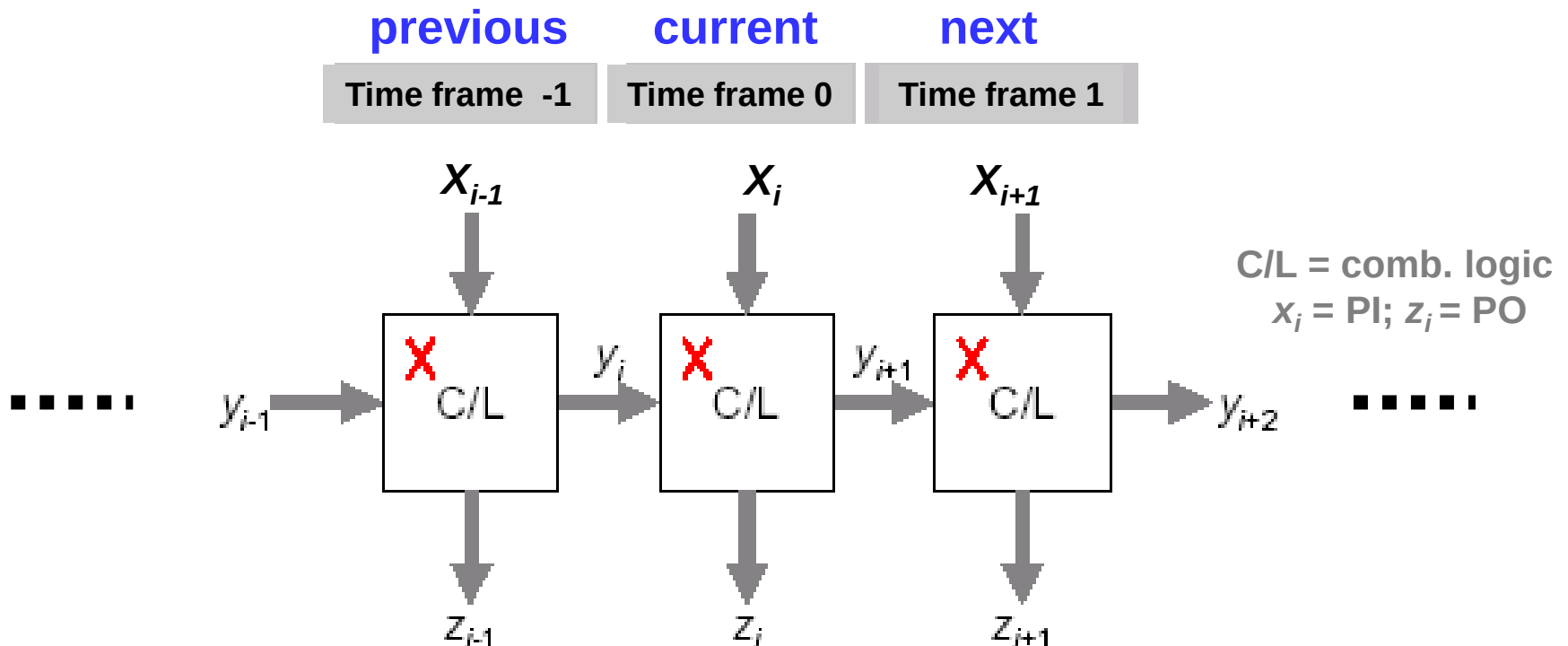
Sequential ATPG

- Introduction
- Time-frame expansion methods
 - ◆ The extended D-algorithm [Kubo 68]
 - ◆ 9-valued D algorithm [Muth 76]
 - ◆ EBT [Marlett 78], BACK [Cheng 88] *
 - ◆ Summary
- Simulation-based methods*
- Issues of Sequential ATPG*
- Conclusions



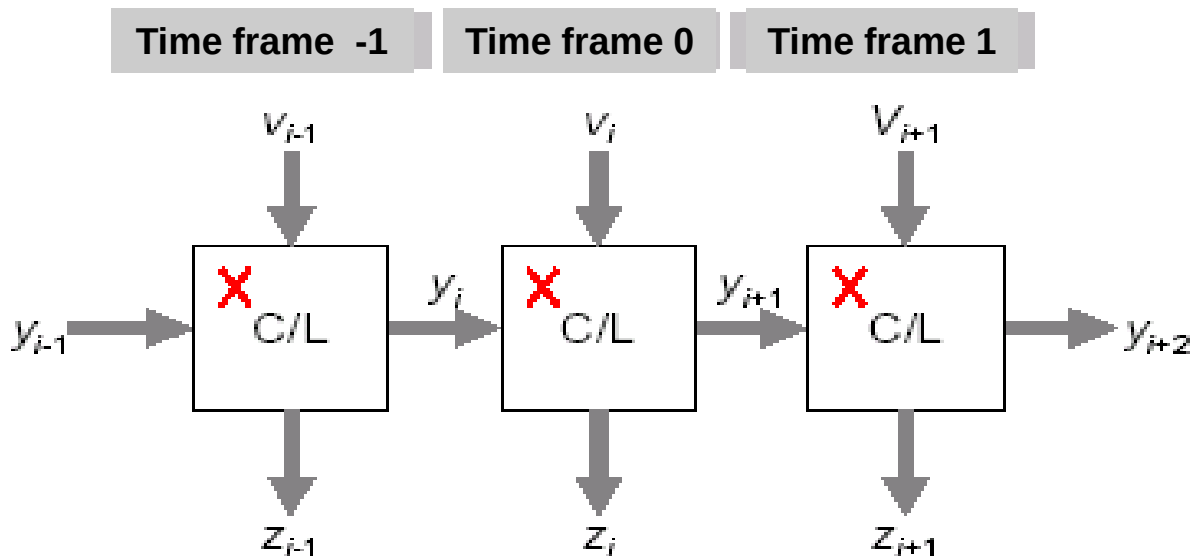
Time Frame Expansion

- IDEA: Replicate circuits and connect time frames **by wires**
 - ◆ y_i = “states”; **No FF!**
 - ◆ Replace **clock cycles** by **space**
- Becomes **combinational ATPG** problem
 - ◆ **NOTE:** Target fault appears in every time frame



Extended D-Algorithm [Kubo 68]

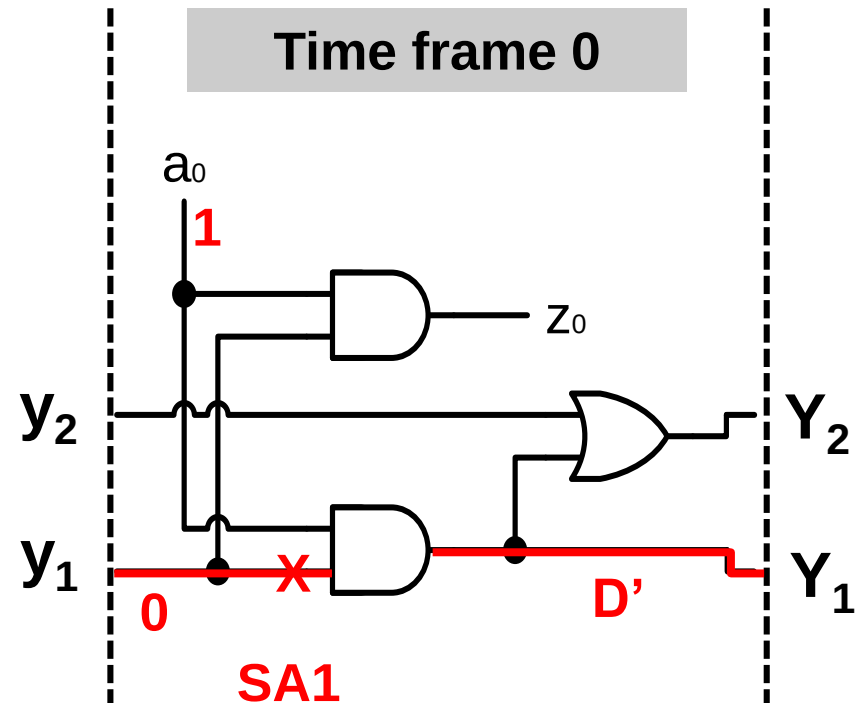
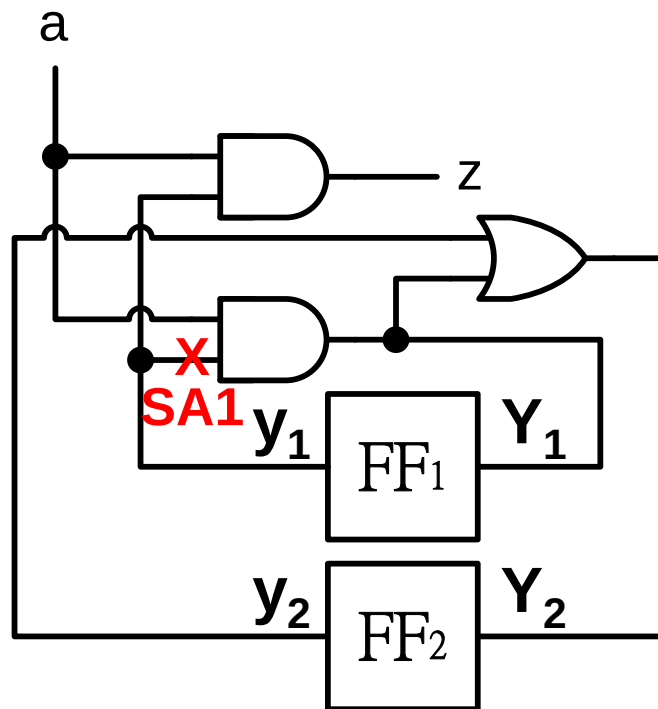
1. Select a target fault f
2. Create a copy of the combinational logic, set it to **time frame 0**
3. Generate a test for f for time frame 0 using D-algorithm
4. If the fault effect is propagated to the FF's, continue fault effect *propagation* in the *next time frame*
5. If there are values required in the FF outputs, continue the *justification* in the *previous time frame*



Example (1)

- STEP 2: create Time frame 0
- STEP 3: generate a test

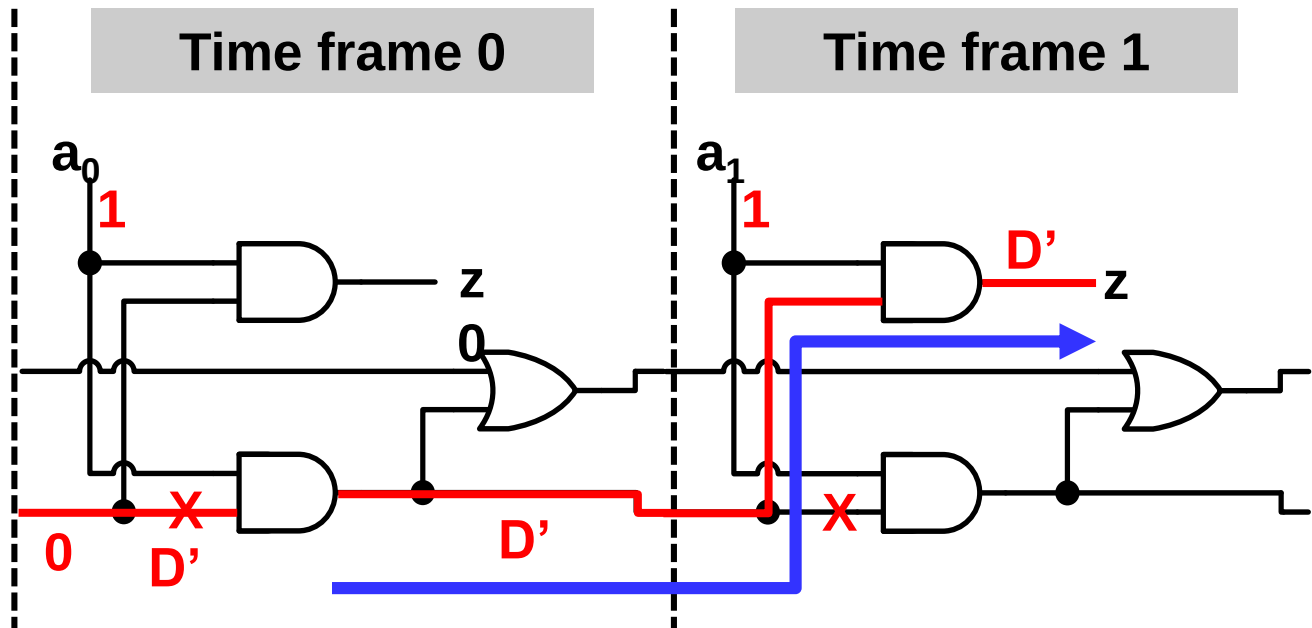
◆ $a_0=1$; $y_1=0$; $Y_1=D'$



Still Need Propagation

Example (2)

- STEP4: Fault effect propagation to time frame 1
 - ◆ $a_1=1$

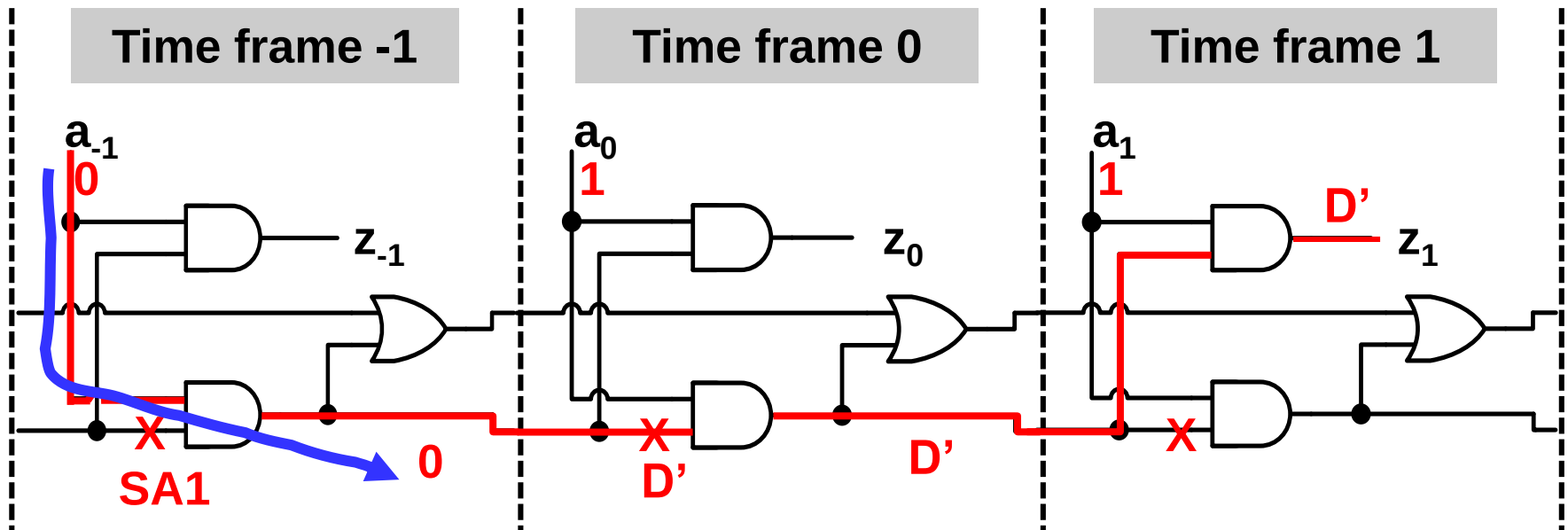


Still Need Activation

Example (3)

- STEP 5: Fault activation back to time frame -1

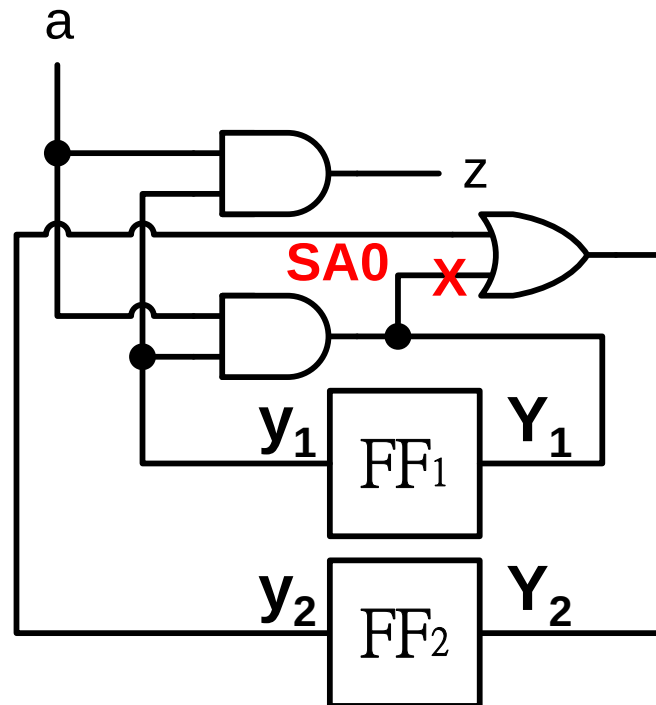
- ◆ $a_{-1}=0$



Test Generated = 0, 1, 1

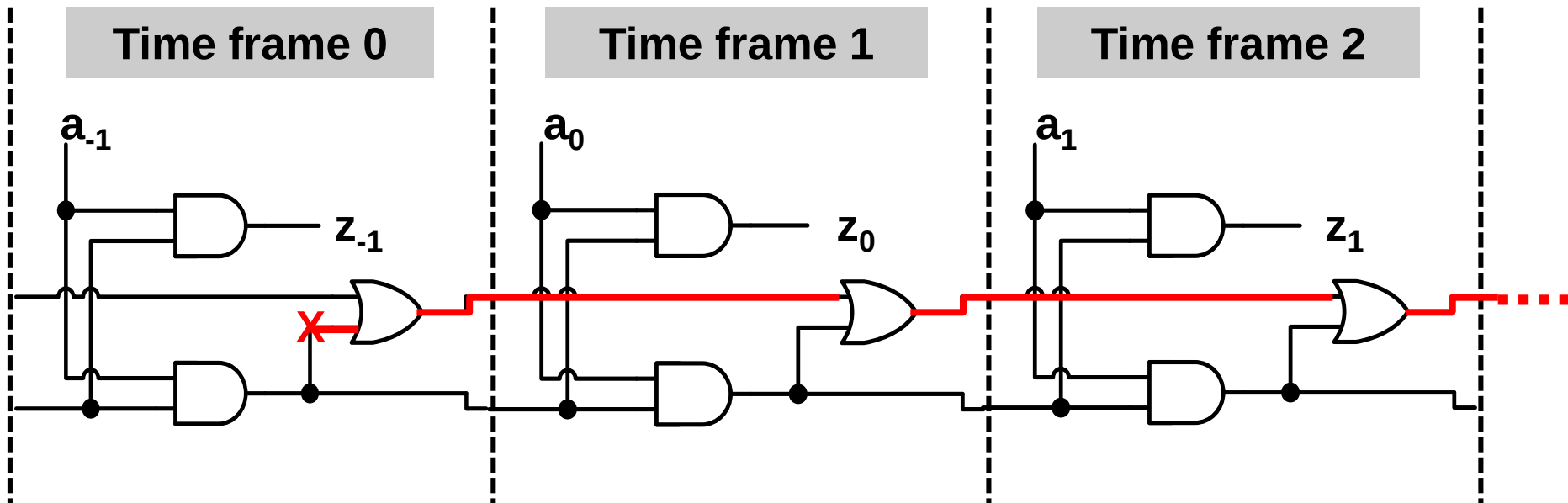
Quiz

Q: Generate a test for stuck-at zero fault in sequential circuit.



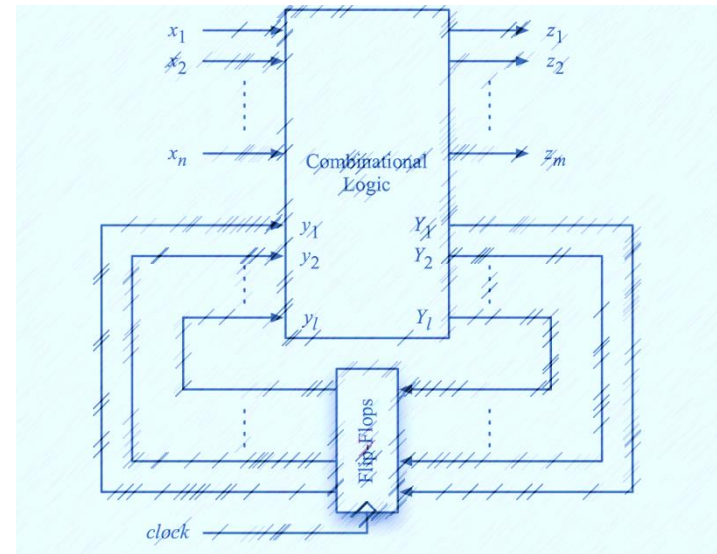
ANS

- No way to propagate
 - ◆ This fault is *untestable* by sequential ATPG
- Endless timeframe expansion ...
 - ◆ Memory explosion!



Summary

- Sequential ATPG
 - ◆ Time and space consuming
 - ◆ Low fault coverage
- Time-frame expansion methods - extended D-algorithm
 - ◆ Replicate circuits into many time frames
 - ◆ Propagate forward, then activated backward
 - ◆ create new time frame if needed



FFT

- In comb. ATPG, first fault **activation**, then **propagation**
- In seq. ATPG, first **propagation**, then **activation**
 - ◆ why?

